

Demonstration of Photonic Chip-based Ultra-high-resolution Dual-band Microwave Imaging of Non-cooperative Targets

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Abstract To break electronic bandwidth bottleneck limited frequency band and resolution, and architectural constraints from generic discrete components, a photonic transceiver chip for radar is presented. It enables concurrent dual-band ultra-high-resolution imaging, demonstrating the first photonic chip-based microwave imaging of non-cooperative targets in realistic environment.

Introduction

Microwave imaging radars are capable of all weather, day-and-night operation, which made them indispensable payloads in remote sensing [1]. Electronic bandwidth bottleneck constrains radar operational modes, such as fixed imaging frequency-band and coarse resolution ($>0.1\text{m}$), whereas photonics provides new design dimensions and abundant bandwidth resource for radar implementation [2,3], and microwave photonics is evolving towards integration for better size, weight, power and cost (SWaP-C) performance and broader application [4]. However, existing photonic integration largely follows conventional radar architecture or inherits microwave photonic links proved by fiber-connected discrete devices. Overemphasis on bandwidth and range resolution fails to obtain clear microwave imagery, suggesting latent phase error in radar signal processing. Furthermore, photonic-chip-based radars have yet to be experimentally validated in realistic scenario involving ambient clutter interference and complex non-cooperative targets comprising distributed scattering centers. Compared to the promised vision, proofs are typically confined to simplified scenarios such as cooperative experiments in darkroom or 1D detection [3,5-7].

In this paper, a photonic transceiver chip for RF processing is proposed and validated through field trials, which also drives architecture evolution of multi-band radar. Supporting wideband signal generation and data-saving stretching reception, the chip enables single synthetic aperture radar system to achieve simultaneous C-band and Ku-band microwave imaging with centimeter-scale resolution in both bands. The radar system yields clear microwave imagery of non-cooperative targets under strong clutter, achieving the first photonic chip-based radar demonstrated in real-world environment.

Chip structure and principle

The chip architecture is illustrated in Fig. 1(a),

with frequency-domain processing depicted in Fig. 1(b). In transmitter (Tx) part, an intermediate-frequency (IF) linearly frequency modulated (LFM) seed sweeping from 3 to 4GHz is modulated onto MZM1, which is biased at maximum transmission point to suppress odd-order modulation sidebands. The generated even-order components in optical domain are subsequently converted into frequency-multiplied dual-band radar waveform E_t , expressed in Eq. (1), at the following photodiode (PD).

$$\begin{aligned} E_t(t) &= E_{t,c}(t) + E_{t,Ku}(t) \\ &= \exp[i(2\pi f_c t + \pi k_c t^2)] \\ &\quad + \exp[i(2\pi f_{Ku} t + \pi k_{Ku} t^2)] \end{aligned} \quad (1)$$

$$f_c = 2f_e, f_{Ku} = 4f_e, k_c = 2k_e, k_{Ku} = 4k_e$$

Here, f_e and k_e are carrier frequency and chirp rate of the input IF seed, and $t \in [-T_p/2, T_p/2]$ is fast time within one radar pulse.

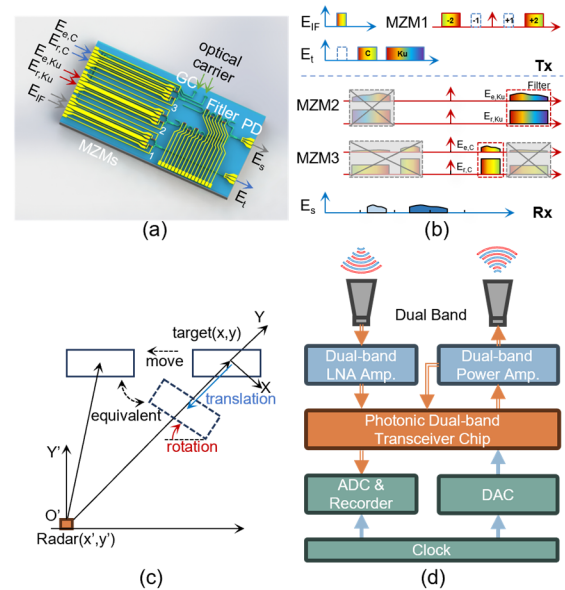


Fig. 1: Principle of photonic chip-based dual-band radar. (a) Chip architecture. (b) Principles of signal processing. (c) Movement of target relative to radar. (d) Photonic chip driven dual-band radar system.

In receiver (Rx) part, for each band, target echo

and reference (replica of generated radar waveform, $E_r=E_i$) are modulated onto the two arms of a shared dual-drive MZM. The cascaded optical filter selects +1st-order sidebands to produce stretched signal E_s upon PD. For inverse synthetic aperture radar (ISAR) imaging shown in Fig. 1(c), target motion $R(\tau) \approx R_t(\tau) + x \cdot \sin[\varphi_r(\tau)] + y \cdot \cos[\varphi_r(\tau)]$ can be decomposed into translational part $R_t(\tau)$ and rotational part $\varphi_r(\tau)=\Omega\tau$, where $\tau \in [0, T_r]$ represents slow time across radar pulses and Ω is effective rotation speed. Target echo is a temporally delayed version of transmitted signal, that is $E_e(t, \tau) = E_{e,C}(t, \tau) + E_{e,Ku}(t, \tau) = \sum_i A_i E_{t,C}[t - 2R_i(\tau)/c] + \sum_j B_j E_{t,Ku}[t - 2R_j(\tau)/c]$, where A_i , B_j are strengths of scattering center i and j at C and Ku band respectively. After motion compensation, the stretched signal is derived as Eq. (2).

$$\begin{aligned} E_s(t, \tau) &= E_{r,C}^*(t) E_{e,C}(t, \tau) + E_{r,Ku}^*(t) E_{e,Ku}(t, \tau) \\ &= \sum_i A_i \exp[i2\pi k_c \cdot 2y_i/c] \\ &\quad \cdot \exp[i2\pi f_c \cdot 2x_i\Omega\tau/c] \\ &\quad + \sum_j B_j \exp[i2\pi k_{Ku} \cdot 2y_j/c] \\ &\quad \cdot \exp[i2\pi f_{Ku} \cdot 2x_j\Omega\tau/c] \end{aligned} \quad (2)$$

The raw data matrix in (t, τ) domain are focused in (f_r, f_d) by further FFT processing, and microwave images can be expressed as Eq. (3).

$$\begin{aligned} E_s(f_r, f_d) &= FFT_{2D}[E_s(t, \tau)] \\ &= \sum_i A_i' \cdot \text{Sinc}[\pi T_p(f_r - k_c \cdot 2y_i/c)] \\ &\quad \cdot \text{Sinc}[\pi T_\tau(f_d - f_c \cdot 2\Omega x_i/c)] \\ &\quad + \sum_j B_j' \cdot \text{Sinc}[\pi T_p(f_r - k_{Ku} \cdot 2y_j/c)] \\ &\quad \cdot \text{Sinc}[\pi T_\tau(f_d - f_{Ku} \cdot 2\Omega x_j/c)] \end{aligned} \quad (3)$$

Owing to distinct chirp rate and carrier frequency between these bands, target (x, y) is mapped to different (f_r, f_d) region, hence dual-band imagery can be separated with reasonable system parameters. Eq. (3) also implies that any intra-pulse and inter-pulse phase errors will result in imaging defocus. As is shown in Fig. 1(d), the dual-band processing capability of the photonic chip enables a single radar platform to perform concurrent dual-band imaging, instead of conventional approach of stacking independent band-specific sub-systems. Moreover, the imaging geometries of the two bands are identical, that is, the images are naturally spatially registered.

Radar signal generation and reception

The photonic chip is designed and fabricated on IMEC iSiPP50G silicon photonics platform. Fig. 2(a) presents a micrograph of the integrated transceiver, packaged in a hermetic module with active thermal control to enhance robustness.

The performance of dual-band signal processing is characterized in Figs. 2(b)–2(d).

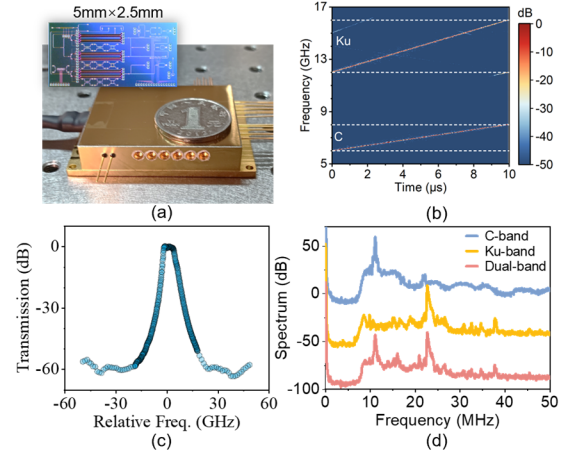


Fig. 2: Signal processing capability of the chip. (a) Micrograph of the photonic TRx chip and picture of the packaged module. (b) Time-frequency analysis of the generated radar waveforms. $T_p=10\mu s$. (c) Transmission response of the optical filter in Rx. (d) Stretched spectrum under B2B test.

The frequency-multiplied waveforms generated by Tx part are captured by a high-speed real-time oscilloscope (Agilent DSOX92004A), and the spectrogram after short-time Fourier transform (STFT) is given in Fig. 2(b). The time-frequency graph confirms simultaneous generation of C-band and Ku-band LFM signals with bandwidths expanded by 2 and 4 times compared with IF input respectively. In addition, the power levels are balanced across both bands, and in-band spurious suppression ratio exceeds 40 dB.

For the receiver, optical filter is very critical to suppress unwanted modulation sidebands and carrier. The measured filter transmission response, obtained via a wavelength tunable laser (Santec TSL-510), is present in Fig. 2(c). The filter exhibits a passband of 6.2 GHz and a steep transition-band slope about 10 dB/GHz. The narrow passband and fast roll-off effectively suppress -1st-order optical mirror component and eliminate harmonic interference in the stretched output. A back-to-back test is conducted by directly connecting the antenna feed cables in Fig. 1(d), and the de-chirped results recorded by electrical spectrum analyzer (Keysight N9030A) are plotted in Fig. 2(d). The de-chirped frequency represents the electrical length within the photonic chip-based radar, which is 11.1 MHz for C-band and 22.7 MHz for Ku-band, consistent with the twofold chirp-rate ratio.

Basic imaging characterization

Imaging quality is quantitatively characterized for both bands through turntable imaging experiment. As is shown in Fig. 3(a), two trihedral corner

reflections (TCRs), which can be regarded as point targets, are placed on a rotary table with an angular rotation rate of 60 deg/s. Spatial resolutions and PSLRs are extracted from slices analysis in both range and azimuth directions. In C-band imagery, 2D resolutions are 7.37cm (range) $\times$ 6.23cm (azimuth), and PSLRs are -15.11 dB and -12.65dB in each dimension. Ku-band shows superior resolution of 4.13cm $\times$ 3.03cm, while the PSLRs in each dimension are -14.96 dB and -12.91 dB.

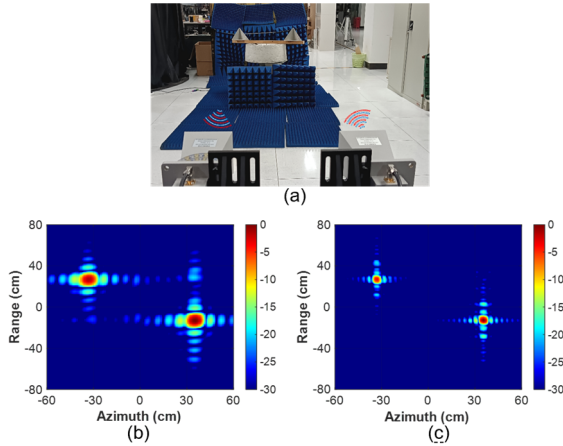


Fig. 3: Characterization of imaging quality achieved by photonic chip-based radar. (a) Scene of turntable experiments. ISAR Imaging results of two TCRs at (b) C-band and (c) Ku-band.

Imaging of non-cooperative targets

The photonic-chip-based radar is deployed at high platform of a building to perform ISAR imaging of passing vehicles. For the scenario in Fig 4(a), clutter from stationary objects, such as trees, terrain and buildings masks target backscattering in the received echoes. The chip's high dynamic range ensures accumulation of enough power from target returns while preventing receiver saturation. After imaging algorithms for correcting range cell migration and phase distortion of target's non-cooperative movement, the dual-band microwave imagery is presented in Fig. 4(c)-(d).

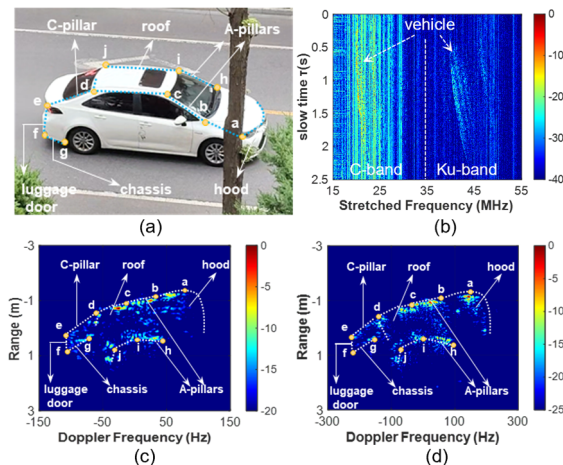


Fig. 4: ISAR imaging of non-cooperative target. (a) Experimental scenario in real-world environment. (b) Interference of clutter on stretched signal. Microwave imagery of a sedan yielded at (c) C-band and (d) Ku-band.

Owing to streamlined design of most vehicles, which comprise many planar facets, the majority of electromagnetic wave is reflected away from the radar. The angular frame structures constitute the dominant scattering centers in microwave imagery. Concurrent dual-band operation and fused radar system intrinsically yields spatially co-registered pictures with coincident imaging projection planes.

Unlike previous contributions that overemphasize bandwidth, which solely determines range resolution, this work addresses comprehensive imaging quality, encompassing two-dimensional resolutions, PSLRs, dynamic range, imaging algorithm and diverse operational band. Consequently, the discrepancy between nominal high resolution and actual insufficient imaging clarity is resolved. To the best of our knowledge, this work contributes the first demonstration of non-cooperative targets imaging in real-world environment using a photonic chip-based radar, with additional features of concurrent dual-band and ultra-high-resolution (centimetre-level).

Conclusion

This work demonstrates a photonic transceiver chip supporting dual-band broadband radar signal generation and reception. The specific design for radar application drives an architecture fusion of dual-band radar system. Its capability of high-fidelity RF processing achieves microwave imaging with resolutions of 7.37cm $\times$ 6.23cm at C-band and 4.13cm $\times$ 3.03cm at Ku-band, both with well 2D PSLRs close to theoretical -13.2dB. To the best of our knowledge, the photonic chip-based radar, for the first time, demonstrates imaging of non-cooperative targets in realistic environment in the field of integrated microwave photonics. Moreover, the photonic chip enabled multi-band concurrent and spatially registered imaging mode will reveal target's frequency-dependent scattering characteristics and obtain target's high-dimensional information for better identification.

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The authors declare no conflicts of interest.

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